Partitions

Lecture notes
M&CS SPbU

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2026

Contents

1 Lecture 1

2 September 2026. Lecture, no seminar.

1.1 Course organization

Course: Lecture + Seminar.

Grading.

- *small version* — Seminar;
- *large version* — Seminar + Final.

Final. Two questions. List of questions about a month in advance.

Schedule.

- Sept 2 — lecture, no seminar;
- Sept 9 — lectures $\times 2$;
- Sept 16 — lecture + seminar (?).

Volunteers, then assigned dates. Dates can be moved.

Contact.

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1.2 Additive number theory

Euler is a large figure in the history of partitions. Motivation: additive number theory.

Definition 1.1 (Basic problem)**.** Let $A = \{a_1, a_2, \dots\}$ be a set of integers. For $n \geqslant 1$, let $A(n)$ be the number of representations of n as a sum of elements of A.

1.3 Goldbach conjecture

Theorem 1.1 (Goldbach conjecture, 1742)**.** *Every even $n > 4$ is a sum of two odd primes:*

$$n = p + q, \qquad p, q \text{ odd primes.}$$

Example 1.1.

$$6 = 3 + 3, \qquad 8 = 5 + 3, \qquad 10 = 7 + 3.$$

1.4 Sums of squares

Definition 1.2. Let $k \geqslant 2$. Let $r_k(n)$ be the number of solutions to

$$n = x_1^2 + \cdots + x_k^2,$$

where each $x_i \in \mathbb{Z} = \{0, \pm 1, \pm 2, \dots\}$ and order matters.

1.4.1 Jacobi's formulae

Jacobi (1829) found simple exact formulae for $r_k(n)$ when $k = 2, 4, 6, 8$.

Example 1.2.

$$r_2(n) = 4\big(d_1(n) - d_3(n)\big),$$

where $d_j(n)$ is the number of positive divisors of n congruent to j modulo 4:

$$d_j(n) = \#\{\, d > 0 : d \mid n,\ d \equiv j \pmod 4 \,\}.$$

Example 1.3 ($n = 5$)**.**

$$\begin{aligned} 5 &= 2^2 + 1^2 = (-2)^2 + 1^2 = 2^2 + (-1)^2 = (-2)^2 + (-1)^2 \\ &= 1^2 + 2^2 = 1^2 + (-2)^2 = (-1)^2 + 2^2 = (-1)^2 + (-2)^2. \end{aligned}$$

So $r_2(5) = 8$. The divisors of 5 are 1 and 5, both $\equiv 1 \pmod 4$, hence $d_1(5) = 2$, $d_3(5) = 0$, and $r_2(5) = 4(2 - 0) = 8$.

1.5 Waring's problem

Definition 1.3 (Waring's problem)**.** Let $k \geqslant 1$ be given. Determine if there is an integer s (depending on k) such that

$$n = x_1^k + x_2^k + \cdots + x_s^k$$

has a solution for every $n \geqslant 1$.

Example 1.4.

- Every $n \geqslant 1$ can be written as a sum of 4 squares.
- Every $n \geqslant 1$ can be written as a sum of 9 cubes.
- Every $n \geqslant 1$ can be written as a sum of 19 fourth powers.

Definition 1.4. Let $g(k)$ be the least such s. Then $g(2) = 4$ (Lagrange, 1770), $g(3) = 9$, $g(4) = 19$.

1.6 Unrestricted partitions

Definition 1.5 (Unrestricted partition)**.** Let $n \geqslant 1$. An *unrestricted partition* of n is a representation of n as a sum of positive integers

$$n = a_1 + a_2 + \cdots + a_k$$

where the order of parts does not matter, k is not restricted, and usually the a_i are written in descending order.

Definition 1.6. $p(n) =$ the number of unrestricted partitions of n. Convention: $p(0) = 1$.

Example 1.5.

n	partitions	$p(n)$
1	1	1
2	2, $1+1$	2
3	3, $2+1$, $1+1+1$	3
4	4, $3+1$, $2+2$, $2+1+1$, $1+1+1+1$	5
5		7
6		11
7		15
$\vdots$		
100		190,569,292

1.7 Hardy–Ramanujan

Theorem 1.2 (Hardy–Ramanujan, 1918)**.**

$$p(n) \sim \frac{1}{4n\sqrt{3}}\, e^{\pi\sqrt{2n/3}} \qquad \text{as } n \to \infty.$$

Definition 1.7. We write $f(n) \sim g(n)$ as $n \to \infty$ if

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = 1.$$

Theorem 1.3 (Euler)**.** *Let $q \in \mathbb{C} \setminus \{0\}$, $|q| < 1$. Then*

$$\begin{aligned}\sum_{n=0}^{\infty} p(n)\, q^n &= 1 + p(1)q + p(2)q^2 + \dots \\ &= 1 + q + 2q^2 + 3q^3 + 5q^4 + \dots \\ &= \prod_{m=1}^{\infty} \frac{1}{1-q^m},\end{aligned}$$

where

$$\prod_{m=1}^{\infty} \frac{1}{1-q^m} = \frac{1}{1-q} \cdot \frac{1}{1-q^2} \cdot \dots$$

Theorem 1.4 (Euler, recurrence)**.**

$$p(n) = \sum_{k\in\mathbb{Z}\setminus\{0\}} (-1)^{k-1}\, p\Big(n - \frac{k(3k-1)}{2}\Big)$$

$$= p(n-1) + p(n-2) - p(n-5) - p(n-7) + \dots,$$

where $p(k) = 0$ for $k < 0$.

Example 1.6.

$$\begin{aligned}p(6) &= p(6-1) + p(6-2) - p(6-5) \\ &= p(5) + p(4) - p(1) = 7 + 5 - 1 = 11.\end{aligned}$$

1.8 Geometric series and the Euler product

Each factor is a geometric series:

$$\begin{aligned}\frac{1}{1-q} &= \sum_{n=0}^{\infty} q^n, \\ \frac{1}{1-q^2} &= \sum_{n=0}^{\infty} q^{2n}, \\ \frac{1}{1-q^k} &= \sum_{n=0}^{\infty} q^{kn} = 1 + q^k + q^{2k} + \dots\end{aligned}$$

The truncated product:

$$
\begin{aligned}
\frac{1}{1-q}\cdot\frac{1}{1-q^2}\cdots\frac{1}{1-q^m} &= \big(1+q+q^{1+1}+q^{1+1+1}+\dots\big)\\
&\qquad\times\big(1+q^2+q^{2+2}+q^{2+2+2}+\dots\big)\\
&\qquad\times\cdots\\
&\qquad\times\big(1+q^m+q^{m+m}+\dots\big)\\
&= \sum_{n_1=0}^{\infty} q^{n_1}\cdot\sum_{n_2=0}^{\infty} q^{2n_2}\cdots\sum_{n_m=0}^{\infty} q^{mn_m}\\
&= \sum_{n_1,n_2,\dots,n_m\geqslant 0} q^{1\cdot n_1+2\cdot n_2+\cdots+m\cdot n_m}.
\end{aligned}
$$

Grouping by total degree,

$$
\begin{aligned}
&= 1+q+\big(q^{1+1}+q^2\big)+\big(q^{1+1+1}+q^{1+2}+q^3\big)+\dots\\
&= \sum_{n=0}^{\infty} p_m(n)\,q^n,
\end{aligned}
$$

where $p_m(n)$ is the number of solutions of $n=a_1+\cdots+a_k$ with $m\geqslant a_1\geqslant\cdots\geqslant a_k\geqslant 1$, i.e. the number of partitions of n into parts not exceeding m.

Theorem 1.5. *Let $m\geqslant 1$. For $q\in\mathbb{C}\setminus\{0\}$, $|q|<1$,*

$$\sum_{n=0}^{\infty} p_m(n)\,q^n=\prod_{k=1}^{m}\frac{1}{1-q^k}.$$

Example 1.7. $p_3(6)=7$:

3+3, 3+2+1, 3+1+1+1, 2+2+2, 2+2+1+1, 2+1+1+1+1, 1+1+1+1+1+1.

1.9 Convergence of infinite products

Definition 1.8. The product $\prod_{n=1}^{\infty}(1+a_n)$ converges if

$$\lim_{N\to\infty}\prod_{n=1}^{N}(1+a_n)$$

exists and is not equal to 0.

Definition 1.9. The product converges absolutely if $\prod_{n=1}^{\infty}(1+|a_n|)$ converges.

Theorem 1.6. *Suppose $a_n>-1$ for all n (or, more generally, $\mathrm{Re}(a_n)>-1$ for all n). Then $\prod_{n=1}^{\infty}(1+a_n)$ converges if and only if $\sum_{n=1}^{\infty}\log(1+a_n)$ converges.*

Theorem 1.7. *The product $\prod_{n=1}^{\infty}(1+a_n)$ converges absolutely iff $\sum_{n=1}^{\infty}a_n$ converges absolutely.*

Example 1.8. $\sum 1/n^2$ converges absolutely, so $\prod_{n=1}^{\infty}\big(1+1/n^2\big)$ converges absolutely. In fact

$$\prod_{n=1}^{\infty}\Big(1+\frac{1}{n^2}\Big)=\frac{e^{\pi}+e^{-\pi}}{2\pi}.$$

1.10 Bounded multiplicity

Definition 1.10. Let $p_{m,d}(n)$ be the number of partitions of n into parts $\leqslant m$ in which each part occurs at most d times.

Theorem 1.8.

$$\sum_{n=0}^{\infty} p_{m,d}(n)\, q^n = \prod_{k=1}^{m} \left(1 + q^k + q^{2k} + \cdots + q^{dk}\right)$$
$$= \prod_{k=1}^{m} \frac{1 - q^{k(d+1)}}{1 - q^k}.$$

1.11 Proof of Euler's generating function

Write

$$F_m(q) := \prod_{k=1}^{m} \frac{1}{1 - q^k}, \qquad F(q) := \prod_{k=1}^{\infty} \frac{1}{1 - q^k}.$$

The product $\prod_{k=1}^{\infty}(1 - q^k)$ converges absolutely for $|q| < 1$, since $\sum q^k$ converges absolutely for $|q| < 1$. Hence $F(q)$ converges for $|q| < 1$.

If $n \leqslant m$, any partition $n = a_1 + \cdots + a_k$ has $a_k \leqslant \cdots \leqslant a_1 \leqslant n \leqslant m$, so $p(n) = p_m(n)$.

Proof. Suppose $0 \leqslant q < 1$. Then

$$F_{m+1}(q) = \frac{1}{1 - q^{m+1}} F_m(q) \geqslant F_m(q),$$

so $\{F_m(q)\}$ is increasing and converges to $F(q)$, and $0 < F_m(q) \leqslant F(q)$.

$$F_m(q) = \sum_{n=0}^{\infty} p_m(n)\, q^n = \sum_{n=0}^{m} p(n)\, q^n + \sum_{n=m+1}^{\infty} p_m(n)\, q^n,$$

hence

$$\sum_{n=0}^{m} p(n)\, q^n \leqslant F_m(q) \leqslant F(q).$$

Thus $\sum p(n) q^n$ converges and

$$\sum_{n=0}^{\infty} p(n)\, q^n \leqslant F(q) = \prod_{k=1}^{\infty} \frac{1}{1 - q^k}.$$

Also $p_m(n) \leqslant p(n)$ for all n, so

$$F_m(q) = \sum p_m(n)\, q^n \leqslant \sum p(n)\, q^n \leqslant F(q).$$

Taking $m \to \infty$,

$$F(q) = \lim_{m\to\infty} F_m(q) \leqslant \sum_{n=0}^{\infty} p(n)\, q^n \leqslant F(q),$$

and therefore

$$\sum_{n=0}^{\infty} p(n)\, q^n = F(q) = \prod_{k=1}^{\infty} \frac{1}{1 - q^k}.$$

This holds for $0 < q < 1$, and extends by analytic continuation to $0 < |q| < 1$. $\square$

1.12 Euler: distinct parts and odd parts

Theorem 1.9 (Euler)**.** *Let $p(D,n)$ be the number of partitions of n into distinct parts, and $p(O,n)$ the number of partitions of n into odd parts. Then $p(D,n) = p(O,n)$ for all n.*

Two types of proof: analytic and combinatorial.

Example 1.9 ($n = 9$)**.** Both equal 8.

distinct parts	odd parts
9	9
$8+1$	$7+1+1$
$7+2$	$5+3+1$
$6+3$	$5+1+1+1+1$
$6+2+1$	$3+3+3$
$5+4$	$3+3+1+1+1$
$5+3+1$	$3+1+1+1+1+1+1$
$4+3+2$	$1+1+1+1+1+1+1+1+1$

Analytic proofs & combinatoric proofs (more fun!).

$$\text{(A)} \qquad \sum_{n=0}^{\infty} p(D,n)\, q^n = \prod_{k=1}^{\infty} (1+q^k), \qquad |q|<1,$$

$$\text{(B)} \qquad \sum_{n=0}^{\infty} p(O,n)\, q^n = \prod_{k=1}^{\infty} \frac{1}{1-q^{2k-1}}, \qquad |q|<1.$$

Analytic proof: transform the RHS of (A) into the RHS of (B).

Proof.

$$\begin{aligned}\prod_{k=1}^{\infty}(1+q^k) &= \prod_{k=1}^{\infty}(1+q^k)\,\frac{1-q^k}{1-q^k} = \prod_{k=1}^{\infty}\frac{1-q^{2k}}{1-q^k} \\ &= \prod_{k=1}^{\infty}\frac{1-q^{2k}}{(1-q^{2k})(1-q^{2k-1})} = \prod_{k=1}^{\infty}\frac{1}{1-q^{2k-1}},\end{aligned}$$

using $\prod_{k=1}^{\infty}(1-q^k) = \prod_{k=1}^{\infty}(1-q^{2k}) \cdot \prod_{k=1}^{\infty}(1-q^{2k-1})$. □

1.13 Graphical representations. Combinatoric proofs

Theorem 1.10 (Euler, 1750)**.** *Let $p_e(D,n)$ be the number of partitions of n into an even number of distinct parts, and $p_o(D,n)$ the number into an odd number of distinct parts. Then*

$$p_e(D,n) - p_o(D,n) = \begin{cases} (-1)^m, & \text{if } n = \dfrac{m(3m\pm 1)}{2}, \\ 0, & \text{otherwise.} \end{cases}$$